

Manav Batavia

Curriculum Vitae

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EDUCATION

Purdue University

Candidate for Ph.D. in Mathematics

Indiana, USA

Aug 2022 - Present

- Current GPA: 4.00/4.00
- Advisors: Prof. Linquan Ma and Prof. Uli Walther

Indian Institute of Technology Bombay

Integrated Master's of Science (B.S.+M.Sc.) in Mathematics

Mumbai, India

Jul 2017 – Jun 2022

- Awarded the Institute Silver medal.
- Cumulative Performance Index (CPI) : 9.82/10.
- Minor in Computer Science and Engineering, Minor CPI: 9.60/10.

PUBLICATIONS AND PREPRINTS

3. **M. Batavia**, K. Mohana Sundaram, T. Murray, V. Pandey, *The arithmetic rank of residual intersections of a complete intersection*, submitted for publication. [[arxiv](#)]
2. H. Ananthnarayan, **M. Batavia**, O. Javadekar, *Syzygies of associated graded modules*, submitted for publication. [[arxiv](#)]
1. S. Banerjee, **M. Batavia**, B. Kane, M. Kyranbay, D. Park, S. Saha, H. C. So, P. Varyani, *Fermat's polygonal number theorem for repeated generalized polygonal numbers*, Journal of Number Theory, Volume 220, 2021, Pages 163-181. [[doi](#)]

INVITED TALKS

The arithmetic rank of residual intersections of a complete intersection <i>Fall Central AMS Sectional</i>	Oct 2025
The arithmetic rank of residual intersections of a complete intersection <i>University of Utah</i>	Sep 2025
The arithmetic rank of residual intersections of a complete intersection <i>Purdue University</i>	Sep 2025
When do associated graded modules have pure resolutions? <i>Poster Presentation, SLMath</i>	Jan 2024

CONFERENCES/WORKSHOPS

AMS Fall Central Sectional Meeting <i>University of St. Louis</i>	Oct 2025
Macaulay2 Workshop <i>University of Wisconsin-Madison</i>	Jul 2025
Singularities in Algebra and Geometry <i>Centro de Investigación en Matemáticas</i>	Jun 2025
Séminaire de Mathématiques Supérieures (SMS) 2025 <i>University of Toronto</i>	Jun 2025
Commutative Algebra in Mixed Characteristic <i>University of Nebraska-Lincoln</i>	Mar 2025
Howard Rowlee Lecture 2025 <i>University of Nebraska-Lincoln</i>	Mar 2025
Cohomologies of Commutative Algebras <i>IIT Madras</i>	Dec 2024
CA+ <i>Iowa State University</i>	Mar 2025
UweFest <i>University of Notre Dame</i>	Aug 2024
Workshop on Recent Trends in Commutative Algebra <i>IIT Bombay</i>	Jun 2024
International Conference on Local Rings and Singularities <i>IIT Bombay</i>	Jun 2024
Upcoming Researchers in Commutative Algebra (URiCA) <i>University of Nebraska-Lincoln</i>	May 2025
Introductory Workshop: Commutative Algebra <i>SLMath, Berkeley</i>	Jan 2024
Connections Workshop: Commutative Algebra <i>SLMath, Berkeley</i>	Jan 2024
Workshop on Commutative Algebra and Algebraic Geometry in Prime Characteristics <i>ICTP, Trieste</i>	May 2023

TECHNICAL SKILLS

Programming Languages: C++, Python, MATLAB.

Softwares: SageMath, Macaulay2, $\text{\LaTeX}$, SolidWorks, AutoCAD, XCircuit.

ACADEMIC ACHIEVEMENTS

- Department Rank 1 in mathematics at IIT Bombay.
- All India rank of 174 in IIT JEE Advanced (out of 0.17 million candidates).
- All India rank of 8 in the B.Math entrance exam for Indian Statistical Institute (ISI).
- Awarded the Mitacs Globalink scholarship for a research internship in Canada.

UNDERGRADUATE RESEARCH EXPERIENCE

Master's Thesis and Further Research

Jul 2021 – Dec 2022

Guide: Prof. Ananthnarayan Hariharan, Department of Mathematics

IIT Bombay

- Proved a necessary and sufficient condition for the existence of a pure resolution of the associated graded module corresponding to a module over a Noetherian local ring.
- Independently proved the Auslander-Buchsbaum-Serre equivalences, and studied Burch's work on constructing ideals with arbitrarily large projective dimension, generated by three or fewer elements.

Betti Tables of Quadratic Ideals in Characteristic Two

Jun - Aug 2021

Guide: Prof. Alexandra Seceleanu, Department of Mathematics

University of Nebraska-Lincoln

This research was done as part of the Polymath Jr. 2021 program.

- Succeeded in resolving all quadratic ideals with three generators in a polynomial ring in three variables over a field of characteristic two. The results attained are easily applicable to larger ideals and larger polynomial rings.
- Presented our results at the Young Mathematicians Conference 2021, organised by the Ohio State University.

Generalization of Fermat's Polygonal Number Theorem

May – Jul 2019

Guide: Prof. Ben Kane, Department of Mathematics

The University of Hong Kong

- Acquired minimal number of terms for which certain weighted sums of generalized polygonal numbers are universal.
- The work done resulted in the aforementioned publication in the Journal of Number Theory.

TEACHING EXPERIENCE

Instructor of Record | *Purdue University*

- Algebra Qualifying Exam Prep Class, Summer 2025
- Applied Calculus I, Fall 2024

Recitation/Tutorial Instructor

- Calculus III | Fall 2023, Spring 2024 and Summer 2024, Purdue University
- Calculus II | Fall 2022 and Spring 2023, Purdue University
- Linear Algebra | Spring 2022, IIT Bombay
- Calculus I | Fall 2019, Fall 2022, IIT Bombay

SERVICE

- Organizer of Ideal Conversations, a student commutative algebra seminar at Purdue University (Aug 2024 - Present)
- Association for Women in Mathematics (AWM) mentor for graduate students at Purdue University (Aug 2023 - Present)
- Department Academic Mentor for undergraduate math students at IIT Bombay (Apr 2019 - May 2022)
- Volunteer for the National Service Scheme (NSS) at IIT Bombay, creating waste awareness and imparting environmental education to underprivileged children (Aug 2017 - May 2018)